Fairness in AI-Assisted Emergency Response Systems

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Abstract

This literature review examines fairness in AI-assisted emergency dispatch and triage, focusing on whether comparable emergency incidents receive comparable response times, dispatch priorities, and transport decisions across demographic groups. Drawing on work in algorithmic fairness, healthcare auditing, and disaster-response AI, we identify a gap in systematic, outcome-level evaluations of operational emergency systems using real-world EMS data. We therefore propose a quantitative audit using public NYC EMS incident data and proxy predictive modeling to evaluate severity-adjusted demographic disparities in emergency response outcomes.

1 Project Overview

The goal of this project is to evaluate fairness in AI-assisted emergency dispatch and triage systems by examining whether response outcomes differ across demographic groups for comparable emergency scenarios. Rather than focusing on aggregate disparities, the analysis emphasizes outcome comparisons within similar clinical contexts — asking whether incidents of comparable urgency receive similar response times, dispatch priorities, and transport decisions across demographic groups.

The project uses the NYC EMS Incident Dispatch dataset, a public city-level administrative dataset containing call types, priority codes, dispatch and response time components, and transport indicators. Because individual-level demographic attributes are unavailable, demographic composition is approximated at the area level using geographic identifiers linked to census data. All analyses are conducted at the area level, with ecological and spatial aggregation limitations explicitly acknowledged.

Methodologically, the project combines descriptive disparity analysis with baseline machine learning proxy models. Incidents are grouped by priority and call characteristics to ensure comparability across cases. Predictive models are trained to estimate response outcomes from incident features, serving as proxies for opaque algorithmic decision support systems. Fairness is evaluated through multiple complementary lenses, including outcome-level fairness, conditional fairness under severity stratification, and group-wise error-rate fairness across demographically defined areas, while distinguishing between operational and algorithmic sources of observed disparities.

2 Related Work

This project builds on prior research in (1) formal algorithmic fairness definitions, (2) auditing bias in public-sector AI systems, and (3) empirical analyses of disparities in healthcare and emergency settings. While fairness in machine learning has been extensively studied in domains such as hiring, lending, and criminal justice, comparatively less work has evaluated fairness in pre-hospital emergency response systems using operational EMS incident data.

2.1 Fairness in Algorithmic Decision-Making

A substantial body of literature formalizes fairness definitions in supervised learning. Barocas et al. [Barocas et al. 2023] provide a comprehensive overview of fairness criteria including demographic parity, equalized odds, and conditional parity, emphasizing that fairness often requires conditioning on relevant contextual features rather than relying on aggregate parity alone. Mehrabi et al. [Mehrabi et al. 2022] survey sources of bias across the machine learning pipeline, demonstrating that disparities can arise from data collection, feature construction, proxy variables, and deployment context — even when protected attributes are excluded from model inputs.

Hardt et al. [Hardt et al. 2016] introduce equality of opportunity, arguing that fairness should be evaluated conditionally by comparing error rates across protected groups among individuals with the same true outcome. This conditional framing directly motivates our evaluation design: emergency incidents are grouped by severity, and both observed outcomes and prediction errors are compared across demographic strata.

Importantly, this literature also emphasizes that fairness can be evaluated through disparities in decision outcomes even when algorithm internals are inaccessible [Barocas et al. 2023; Mehrabi et al. 2022]. This insight is particularly relevant in public-sector contexts, where proprietary or embedded decision-support systems limit direct access to model structure and parameters.

While these works establish foundational fairness definitions and auditing frameworks, they do not apply conditional fairness evaluation to operational emergency dispatch systems using real-world, public EMS incident-level data. Nor do they combine outcome-level disparity analysis with proxy modeling in a pre-hospital emergency response setting. Our project extends these theoretical frameworks to a high-stakes public-sector domain by operationalizing formal fairness within severity-stratified EMS data and evaluating both observed response outcomes and proxy-model behavior.

2.2 Algorithmic Fairness in Public-Sector Systems

Several studies have examined fairness in public-sector decision-making systems, including healthcare and social services. In healthcare, Obermeyer et al. [Obermeyer et al. 2019] identify racial bias in a widely used risk prediction algorithm caused by using healthcare cost as a proxy for illness severity, illustrating how outcome-level auditing can reveal disparities even when protected attributes are excluded. This study demonstrates the risks of proxy variables and informs our evaluation strategy; however, it focuses on longitudinal healthcare risk scoring rather than time-critical emergency dispatch and response systems.

More recent work has examined bias in AI systems used explicitly for emergency decision-making. Adam et al. [Adam et al. 2022]

analyze how biased artificial intelligence can influence emergency decisions and propose ethical, operational, and governance-oriented mitigation strategies. Similarly, Gevaert et al. [Gevaert et al. 2021] and Visave [Visave 2024] emphasize fairness, accountability, and transparency in disaster and emergency AI systems embedded in complex sociotechnical environments. However, these studies are primarily conceptual and do not conduct systematic, quantitative evaluations of demographic disparities using real-world EMS incident data or formal conditional fairness metrics.

2.3 Empirical Disparities in Emergency and Healthcare Data

Research using EMS and emergency-care data has traditionally emphasized operational efficiency — such as response time optimization, triage flow, and geographic access — rather than formal fairness evaluation. These studies often document substantial variation across neighborhoods or service areas. However, many do not frame disparities using formal algorithmic fairness definitions, nor do they consistently condition comparisons on incident severity or call characteristics. As a result, it can be difficult to distinguish operational constraints from potential inequities.

More recent work has examined inequities in emergency-care outcomes using rigorous quantitative methods. Wang et al. [Wang et al. 2024] analyze racial and ethnic disparities in emergency department wait times using a large single-center dataset and a Blinder–Oaxaca decomposition approach. They find that a substantial portion of observed disparities can be explained by measurable factors such as triage acuity and arrival mode, while a nontrivial fraction remains unexplained. This study is highly relevant because it uses real operational healthcare data and conducts conditional analysis focused on outcome-level disparities. While methodologically rigorous, the study does not situate its analysis within formal ML fairness definitions, nor does it evaluate algorithm-supported decision systems or employ proxy modeling to audit opaque decision processes.

Complementing real-data disparity analyses, Bahamazava [Bahamazava 2025] models algorithmic bias in AI-driven emergency response systems using simulations grounded in economic theory. The study demonstrates how biased allocation rules can lead to delayed responses and worse outcomes for disadvantaged groups. While this work highlights the systemic consequences of biased algorithms, it relies on simulated environments rather than real operational EMS incident data.

Kozcuer et al. [Kozcuer et al. 2024] propose a transnational fairness framework for disaster-response machine learning systems and demonstrate performance disparities across socio-economic country groupings using image classification tasks. Their work strengthens the case for explicitly defining demographic groupings and contextual fairness criteria in emergency-related AI applications. However, the empirical setting differs from operational dispatch or triage systems, as it focuses on classification performance rather than response-time or resource-allocation outcomes.

Existing empirical studies either analyze disparities without explicitly connecting to formal ML fairness frameworks (Wang et al.), rely on simulation rather than operational data (Bahamazava), or evaluate classification models rather than real-time dispatch and response outcomes (Kozcuer et al.). To our knowledge, no prior study combines real EMS incident data, structured comparisons across comparable emergency contexts, and proxy modeling within a single evaluation framework. Our project aims to integrate these elements into a systematic, outcome-level fairness evaluation of AI-assisted emergency response.

2.4 Evaluating Fairness with Proxy Models

When deployed AI systems are proprietary or not directly observable, one practical strategy is to train proxy predictive models on observed inputs and outputs to approximate algorithmic decision behavior [Barocas et al. 2023; Hardt et al. 2016]. This approach enables quantitative auditing of fairness using standard supervised learning tools.

Specifically, proxy modeling allows comparison of (i) observed outcome disparities, (ii) model-predicted outcome disparities, and (iii) group-wise model error rates within comparable emergency conditions. Such distinctions help clarify whether disparities arise from operational processes, from predictive modeling behavior, or from differential error patterns across demographic groups.

Proxy modeling has been used to audit hiring systems, risk assessment tools, and other opaque decision-support systems. However, it has not been systematically applied to public EMS dispatch data to evaluate conditional fairness in emergency response timing, prioritization, and transport decisions. This gap motivates our use of proxy modeling to audit AI-assisted emergency response systems using real-world incident-level data.

2.5 Gap and Contribution

Despite growing interest in fairness in public-sector AI, there remains a gap in systematic, outcome-level evaluations of fairness in AI-assisted emergency response systems that combine (i) severity-stratified outcome comparisons and (ii) proxy-model auditing of opaque decision systems. Existing work has either emphasized conceptual and ethical considerations [Adam et al. 2022; Gevaert et al. 2021; Visave 2024], relied on simulated settings [Bahamazava 2025], or examined real-world disparities without explicitly integrating formal machine learning fairness definitions and auditing practices [Wang et al. 2024].

Our analysis builds on these insights by applying conditional fairness concepts and proxy predictive modeling to operational EMS incident data. By stratifying incidents by severity and comparing both observed outcomes and proxy-model behavior across demographic groups, the analysis moves beyond descriptive disparities toward a quantitative fairness evaluation in a high-impact domain where algorithmic systems may influence time-critical decisions.

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